# BENZOYL PEROXIDE

Bleaches fabric & hair

No bacterial resistance

**CHEMICAL NAME** Dibenzoyl peroxide (C14H10O4)
**CLASS** Topical antimicrobial and keratolytic (acne)
**TRADE NAMES** PanOxyl · Acnecide · Benzac AC · Epiduo (with adapalene)

## THERAPEUTIC USES / INDICATIONS

- Mild to moderate acne vulgaris (alone or with topical retinoid/antibiotic).
- Used with topical/oral antibiotics to prevent antibiotic resistance.

## PHARMACODYNAMICS

- Releases **free oxygen radicals** → bactericidal against *Cutibacterium acnes* (no resistance develops); keratolytic and comedolytic; reduces inflammation.

## ADMINISTRATION

- **Topical** gel, wash or cream to whole affected area (not just spots) once or twice daily after washing.
- Start with low strength (2.5–5%) once daily, increase as tolerated.
- Avoid eyes, mouth, mucous membranes.
- Can bleach hair, clothing, towels, bedding.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Dryness, peeling, redness, burning (first weeks).
- Bleaching of hair and fabrics.
- Stinging.

### ADVERSE EFFECTS (SERIOUS)

- Contact dermatitis (allergic).
- Severe irritation; rare hypersensitivity (throat tightness).

## DOSAGES

- 2.5–10% applied once or twice daily.
- Response in 6–8 weeks; continue for maintenance.

## CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Broken or eczematous skin (relative).

### KEY INTERACTIONS

- **Topical tretinoin (some formulations):** may be oxidised — apply at different times (morning/evening).
- Other irritant products ↑ dryness.

## PHARMACOKINETICS

- ~5% absorbed through skin; converted to benzoic acid and excreted in urine.

## NURSING MANAGEMENT

- Assess acne severity; check for scarring (refer if severe).
- Teach application and sun protection.
- Manage irritation (reduce frequency, moisturise).

## PATIENT EDUCATION

- Apply a thin layer to the whole area, not only spots.
- Expect dryness at first; use a non-comedogenic moisturiser.
- Use white towels and pillowcases — it bleaches fabrics.
- Results take 6–8 weeks.

**KEY POINTS**

- Bactericidal + keratolytic for acne.
- Dryness, irritation.
- No resistance; pair with antibiotics.
- Bleaches fabrics and hair.

**SOURCES CHECKED** **[975]** Adapalene / Benzoyl Peroxide Zentiva 0.1%/2.5% gel (UK SmPC)

*Full references p. 432–457*